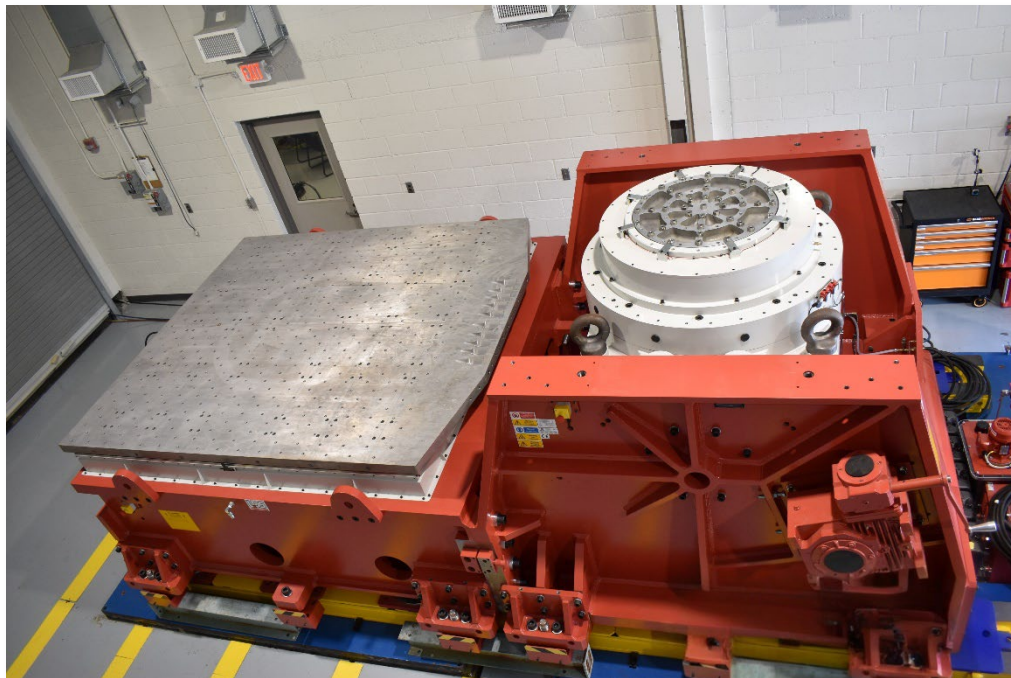




When large payloads need high performance vibration or shock testing, MSFC's East Vibration Test Lab gives engineers the confidence they need to design and test highly reliable products. This system has been configured in single and multi-shaker setups, and it has been used to test products such as spacecrafts, satellites, and missiles.

Characteristics

Test Article Support Capacity	11,023 lb.
Shaker Body Mass (M_b)	28,590 lb.
Stray Magnetic Field*	20 gauss
Low Gauss Option	3 gauss
Slip Table Size	98in x 98in
Rated Amplifier Output	300 kVA
384K Amplifier Weight	2,695 lb.

*Stray magnetic field is theoretical maximum at normal operating temperature.

Industry Applications

- 3-axis testing of integrated system-level hardware in accordance with SMC-S-016 and NASA-STD-7002 standards.
- Supporting NASA, Department of Defense, and commercial industry development, qualification, and acceptance testing.
- Optional clean room environment and access to high bay bridge cranes.

Performance Parameters

Armature Diameter	29.9 in
Sine Force (peak)	67,500 lbf
Random Force (rms)	60,000 lbf
Maximum 1/2-sine Shock Force	180,000 lbf
Armature Resonance (f_n)	1,325 Hz
Usable Frequency Range	DC to 1.7kHz
Velocity (sine peak) – full-field	74.8 in/s
Acceleration (sine, peak)	75 g
Acceleration (random, rms)	60 g
Displacement (cont. pk-pk)	2.0 in



Features

- Combination of high-performance armature design and water-cooled coils deliver excellent acceleration and velocity performance.
- Automatic armature and body position load compensation system ensures larger loads can be comfortably accommodated.
- Trunnions feature Lin-E-Air suspension system which allows the vibrator body to be suspended on air bags and guided on shafts.

